Experiment-5

Nmap host discovery in tryhackme platform

Aim:

To use nmap to discover live hosts using ping scan techniques in the tryhackme

Requirements:

Tryhackme acc (free), Attack box, nmap install

Procedure:

1) connect to tryhackme and start the target room machine

2) Launch attack box, or connect via open vpn

3) ARP Scan : nmap -PR -sn <target-IP_range> - view : MAC addresses, and vendors

4) ICMP echo scan : nmap -PE -sn <target-IP> observe

5) ICMP Timestamp : nmap -PP -sn <target-IP>

6) TCP SYN Ping on port 80: nmap -PS80 -sn <target-IP>

7) TCP SYN Ping on multiple ports : nmap -PS22,80,443 -sn <target-IP>

commands

ARP scan

nmap -PR -sn 192.168.1.0/24

ICMP echo ping scan

nmap -PE -sn 10.10.x.x

ICMP Timestamp scan

nmap -PP -sn 10.10.x.x

TCP SYN Ping (common ports)

nmap -PS22,80,443,8080 -sn 10.10.x.x

TCP ACK Ping

nmap -PA80 -sn 10.10.x.x

UDP ping

nmap -PU53,161 -sn 10.10.x.x

All scan types combined + verbose

nmap -PE -PS22,80 -PA443 -PU53 -sn -v
10.10.x.x/24

show reason for host status

nmap --reason -sn 10.10.x.x

save output to file

nmap -sn -oN discovery.txt 10.10.x.x/24

8) UDP ping: nmap -PU53 -sn <target-IP>.

9) combined scan with verbose: nmap -PE -PS22,80,PU53 -sn -v <range>

10) full discovery report: nmap --reason -sn <target-IP> to see detection reason

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Result:

various nmap host discovery techniques were successfully executed in the tryhackme environment. live hosts were identified using ARP, ICMP, TCP, and UDP probes. Differences in detection reliability were observed